Agnostic oracle bounds for Bayesian neural networks

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Abstract

We study whether Bayesian neural-network prediction can compete with the best small network when the data need not be generated by any network. Labels are bounded, inputs range over a continuous domain, and training observations are independent and identically distributed. We give two guarantees with coefficient one on the comparator's population squared loss. First, for a fully connected Gaussian tanh prior in the strong-prior limit, we improve a tempered-likelihood oracle bound by retaining the learner's energy penalty in the statistical comparison. At fixed depth $L \geq 3$, the remainder decreases as $\widetilde{O}(n^{-1/(L-1)})$, instead of $\widetilde{O}(n^{-1/(L+1)})$, at every learner width at least the teacher width; one hidden layer gives a square-root rate. Related bounds hold at finite Gaussian prior scale for every $\gamma \geq 2$, and for every $\gamma \geq 1$ with one hidden layer. Second, a prior that explicitly mixes over network architectures gives a final-posterior-mean bound in expectation of order the teacher parameter count divided by n, with additional logarithmic and polynomial-depth factors. This second result covers tanh, ReLU and GELU, but changes the prior. It uses a continuous-prior version of a recent exponential-weights aggregation theorem, for which we provide a short proof. We distinguish these results from exact interpolation, width-only comparison with arbitrarily large teacher weights, and fast high-probability guarantees for Bayesian model averaging.

1 The approximation question

Real data need not have a small exact neural implementation. A useful learning guarantee should instead compare a predictor with the best small network that approximates the data. Let D be any distribution on $X \times [-1,1]$, with $X \subseteq [-1,1]^d$, and let

$$R(f) = \mathbb{E}_{(x,y)\sim D}(f(x)-y)^2, \qquad \widehat{R}_S(f) = \frac{1}{n}\sum_{i=1}^{n}(f(x_i)-y_i)^2.$$

We seek bounds of the form

$$R(\widehat{f}_S) \leq R(g) + \text{a complexity penalty tending to zero},$$

where g is a small multilayer perceptron and the learner may have arbitrarily larger width. Such a bound is *agnostic*: it makes no realizability assumption, and permits arbitrary bounded noise and model misspecification. An additive remainder is necessary even for estimating an unknown constant regression mean.

There are two different sources of a favorable prior. One can start with independent Gaussian weights in a fixed, possibly very wide architecture and prove that their induced preferences are useful. Alternatively, one can explicitly assign prior mass to small architectures. We treat both approaches, and keep their conclusions separate.

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Gaussian weights and a tempered likelihood. For tanh networks with $L-1$ hidden layers, our first theorem uses exactly the Gaussian energy and raw-output likelihood defined in Section 2. The prior scale tends to zero after the sample and learner width are fixed. A previous proof [1] bounded the complexity of every empirical minimizer using only its maximum possible energy. We instead retain the negative learner-energy term when comparing with a teacher. Statistical deviations can then be absorbed by this term. The resulting bound holds with high probability on one event for every eligible learner width, for posterior draws in risk expectation and for the mean of clipped predictions. The raw-output extension in Appendix A uses a different calibration and a standard smooth-loss theorem. Section 3 also gives finite-prior versions, without a prior-strength limit. The depth-dependent sample exponent remains a substantial limitation.

An explicit prior over architectures. Our second theorem uses a fixed Gaussian observation likelihood centered on clipped predictions and a prior that chooses a network size, then continuous weights. It gives a bound for the final posterior mean using all observations, with no zero-temperature limit or averaging over training prefixes. At fixed input dimension and depth, a width-m teacher with polynomially bounded weights has an expected excess bound $\widetilde{O}(m^2/n)$. Consequently the learner can compete with widths $m \le n^\alpha$ for every fixed $\alpha < 1/2$. This is a stronger approximation guarantee, but it builds the preference for small architectures into the prior. It does not establish the same bound for independent Gaussian weights on every neuron of a large fixed architecture.

Relation to aggregation theory. Høgsgaard, Rebeschini and Wegel [2] established the sharp expected-risk behavior of final exponential-weights aggregation under random design, resolving a question about this estimator at sufficiently high constant temperature. The statistical principle behind our second theorem belongs to this line of work. We give a self-contained continuous-prior argument, with a conservative temperature constant, and a quantitative neural-network application. Fast expectation bounds do not automatically imply fast bounds with high probability.

2 A sharper oracle inequality for a Gaussian weight prior

This section concerns a fully connected network with $H \ge 1$ hidden layers, all of width N, and a scalar linear readout. Its total depth, counting the readout, is $L = H + 1$. The activation in every hidden layer is tanh. Writing $\bar{x} = (1, x) \in \mathbb{R}^{d+1}$, the network is

$$h^1(x) = \tanh(W^1 \bar{x}), \qquad h^\ell(x) = \tanh(W^\ell h^{\ell-1}(x)) \quad (2 \le \ell \le H), \qquad z_w(x) = v^\top h^H(x).$$

The activation is applied coordinatewise. There are no internal biases; the constant input coordinate supplies the first-layer biases. Define the weighted quadratic energy

$$Q_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{\ell=2}^{H} \|W^\ell\|_F^2 + N\|v\|_2^2. \tag{1}$$

Here and below $d \ge 1$. Inputs may have any distribution on $\mathbb{R}^d$; the labels satisfy $|Y| \le 1$. No bound on the input norm is needed for the argument in this section.

Let $S = ((X_i, Y_i))_{i=1}^n$ consist of independent samples from a law D. For a predictor g, write

$$R(g) = \mathbb{E}_D(g(X) - Y)^2, \qquad \widehat{R}_S(g) = \frac{1}{n}\sum_{i=1}^{n}(g(X_i) - Y_i)^2, \qquad \mathrm{clip}(t) = \max\{-1, \min\{1, t\}\}.$$

For positive σ and λ, give the weights independent centered Gaussian priors with variances

$$\mathrm{Var}(W^1_{jk}) = \frac{N\sigma^2}{d}, \qquad \mathrm{Var}(W^\ell_{jk}) = \sigma^2 \ (2 \le \ell \le H), \qquad \mathrm{Var}(v_j) = \frac{\sigma^2}{N}. \tag{2}$$

Use a Gaussian likelihood with mean $z_w(X_i)$ and variance $n\lambda\sigma^2$ for each observation. The posterior is therefore

$$\Pi_{\sigma,N,S}(dw) = \frac{1}{Z_{\sigma,N,S}} \exp\left[-\frac{Q_N(w) + \lambda^{-1}\widehat{R}_S(z_w)}{2\sigma^2}\right] dw. \tag{3}$$

The likelihood uses the raw real-valued network output; it does not require exact interpolation. The limit $\sigma \downarrow 0$ is taken with n, S, N, and λ fixed. Its limiting optimization problem is

$$\underset{w}{\mathrm{minimize}}\{\widehat{R}_S(z_w) + \lambda Q_N(w)\}. \tag{4}$$

2.1 Teacher implementations and a width-independent capacity bound

Lemma 2.1 (Cloning a teacher)**.** *Suppose that g is a network of the preceding architecture, hidden width m, and depth $H+1$, with every weight of absolute value at most B. At every width $N \ge m$ it has an exact implementation with*

$$Q_N \le E := d(d+1)B^2 + 2Hm^2B^2. \tag{5}$$

Moreover, $|g(x)| \le U := mB$ for every x.

Proof. Partition the N neurons in each hidden layer into m groups, of sizes $k_j \in \{\lfloor N/m \rfloor, \lceil N/m \rceil\}$. Copy a teacher first-layer row onto every neuron in its group. Replace a teacher internal weight A^ℓ_{ji} by A^ℓ_{ji}/k_i on every corresponding edge, and its readout weight u_j by u_j/k_j on each copy. Every copy has exactly the teacher activation, so the function is unchanged. The first-layer energy is at most $d(d+1)B^2$. Since $k_j/k_i \le 2$, each internal layer costs at most $2m^2B^2$. Since $k_j \ge N/(2m)$, the readout costs at most $2m^2B^2$. There are $H-1$ internal layers and one readout. Finally, $|\tanh| \le 1$ gives $|g| \le \sum_j |u_j| \le mB$. □

Put

$$D_n = \sqrt{2(d+1)\log(n+1)} + \frac{H-1}{2}, \qquad \mathcal{Z}_K = \bigcup_{N \ge 1} \{z_w : Q_N(w) \le K\}. \tag{6}$$

The union over all learner widths is essential: the following event controls every width at once.

Lemma 2.2 (Capacity of an energy ball)**.** *For every set of n inputs and every $K > 0$, the empirical Rademacher complexity satisfies*

$$\widehat{\mathfrak{R}}_S(\mathcal{Z}_K) \le \left(\frac{K}{H}\right)^{H/2} \frac{D_n}{\sqrt{n}}. \tag{7}$$

Consequently, with probability at least $1-\eta$, simultaneously for all widths and all w with $Q_N(w) \le K$,

$$R(\mathrm{clip}\, z_w) \le \widehat{R}_S(\mathrm{clip}\, z_w) + 8\left(\frac{K}{H}\right)^{H/2} \frac{D_n}{\sqrt{n}} + 4\sqrt{\frac{\log(1/\eta)}{2n}}. \tag{8}$$

Proof. For a bounded set $A \subset \mathbb{R}^n$, write $\mathrm{Rad}(A) = n^{-1}\mathbb{E}_\epsilon \sup_{a\in A} \sum_i \epsilon_i a_i$. Its value is unchanged by taking a closed convex hull; it scales linearly under positive scalar multiplication. We use the standard Rademacher contraction inequality, Massart's finite-class bound, and the Sauer–Shelah bound for halfspaces.

For $t \in [0,1]$, let $\mathrm{clip}_t(u) = \max\{-t, \min\{t, u\}\}$, and let $\mathrm{Clip}(A)$ contain the vectors $(\mathrm{clip}_t(a_i))_{i=1}^n$ for $a \in A$ and a single common $t \in [0,1]$. The identity $\mathrm{clip}_t(u) = (|u+t| - |u-t|)/2$, contraction, and symmetry of the Rademacher signs imply

$$\mathrm{Rad}(\mathrm{Clip}(A)) \le \mathrm{Rad}(A) + \frac{1}{2\sqrt{n}}. \tag{9}$$

Indeed, split the supremum between the two absolute-value expressions and contract each. The two additional terms sum to $\mathbb{E}|\sum_i \epsilon_i|/(2n) \le 1/(2\sqrt{n})$.

We also need a positive mixture identity. For $s > 0$ and $|u| \le 1$,

$$\frac{\tanh(su)}{s} = \int_0^1 \mathrm{clip}_t(u)\, 2s \tanh(st)\, \mathrm{sech}^2(st)\, dt + \mathrm{sech}^2(s)\, \mathrm{clip}_1(u). \tag{10}$$

All weights are nonnegative and have total mass one. To verify the identity for $u \ge 0$, integrate by parts with $\psi(t) = \tanh^2(st)$: the right side is $u - \int_0^u \tanh^2(st)dt = \tanh(su)/s$. Oddness gives the other half.

On the fixed sample inputs, let $\mathcal{F}_1$ be the closed convex hull of the vectors $(\tanh(a \cdot \bar{X}_i))_{i=1}^n$, with $a \in \mathbb{R}^{d+1}$. Recursively set $\mathcal{F}_\ell = \overline{\mathrm{conv}}\,\mathrm{Clip}(\mathcal{F}_{\ell-1})$. These sets are symmetric, convex, contain zero, and lie in $[-1,1]^n$. Define $\rho^1 = \mathbf{1}_N$ and $\rho^\ell = |W^\ell|\rho^{\ell-1}$, with entrywise absolute values. Induction using (10) shows that the sample vector of hidden neuron j at layer ℓ belongs to $\rho_j^\ell \mathcal{F}_\ell$. A zero value of ρ_j^ℓ forces the neuron to vanish. For a positive value, divide the preactivation by ρ_j^ℓ; it is a convex signed combination of the previous normalized neurons, hence belongs to $\mathcal{F}_{\ell-1}$. The mixture identity completes the induction.

It follows that the output sample vector belongs to $P(w)\mathcal{F}_H$, where $P(w) = |v|^\top |W^H| \cdots |W^2| \mathbf{1}_N$ (with no matrix factors when $H = 1$). Cauchy–Schwarz and the Frobenius bound on operator norm give

$$P(w) \le \sqrt{N}\|v\|_2 \prod_{\ell=2}^H \|W^\ell\|_F \le \left(\frac{Q_N(w)}{H}\right)^{H/2}.$$

The last inequality is AM–GM over the H energy terms after the first layer. The first-layer energy is not needed for this capacity estimate. By (9), $\mathrm{Rad}(\mathcal{F}_H) \le \mathrm{Rad}(\mathcal{F}_1) + (H-1)/(2\sqrt{n})$.

Finally, if a scalar random variable T has distribution function $(1+\tanh t)/2$, then $\tanh u = \mathbb{E}\,\mathrm{sign}(u - T)$. Thus $\mathcal{F}_1$ lies in the convex hull of the affine-halfspace dichotomies of the sample inputs. There are at most $(n+1)^{d+1}$ such dichotomies by Sauer–Shelah. Massart's bound gives $\mathrm{Rad}(\mathcal{F}_1) \le \sqrt{2(d+1)\log(n+1)/n}$, proving (7).

The clipped squared loss is bounded by 4 and is 4-Lipschitz in the raw prediction. Symmetrization and contraction bound its expected uniform deviation by eight times (7). Changing one observation changes that supremum by at most $4/n$, so bounded-difference concentration gives (8). The suprema are measurable: for each width the relevant loss is continuous in the parameters, and rational parameter vectors in the open energy ellipsoid are dense in the closed ellipsoid. Taking the union over the countably many widths gives a countable determining class. □

2.2 Absorbing capacity into the energy penalty

An estimate based only on the largest possible energy of a minimizer loses useful information. In the proof below, the capacity cost of the selected network is paid by that same network's energy

penalty.

Theorem 2.3 (Gaussian oracle inequality, at least two hidden layers)**.** *Suppose $H \geq 2$, $0 < \delta < 1$, and $n \geq 1$. Set*

$$a_n = 8(2/H)^{H/2}\frac{D_n}{\sqrt{n}}, \qquad \lambda = a_n^{2/H} = \frac{2}{H}\left(\frac{8D_n}{\sqrt{n}}\right)^{2/H}, \tag{11}$$

$$J = \max\{0, \lceil \log_2(1/\lambda)\rceil\}, \qquad b_n = 4\sqrt{\frac{\log(2(J+1)/\delta)}{2n}}, \qquad e_n = 8H^{-H/2}\frac{D_n}{\sqrt{n}}. \tag{12}$$

Fix a teacher g as in Lemma 2.1, independently of the observed sample. With probability at least $1-\delta$, simultaneously for every learner width $N \geq m$, every global minimizer w of (4) satisfies

$$R(\operatorname{clip} z_w) \leq R(g) + \lambda E + e_n + b_n + h_g, \tag{13}$$

where one may use either of the following teacher concentration bounds, chosen in advance (each choice gives its own event of probability at least $1-\delta$):

$$h_g = (U+1)^2\sqrt{\frac{\log(2/\delta)}{2n}}, \tag{14}$$

$$h_g = (U+1)\sqrt{\frac{2R(g)\log(2/\delta)}{n}} + \frac{2(U+1)^2\log(2/\delta)}{3n}. \tag{15}$$

On that same event, for each fixed $N \geq m$, the right side of (13) also bounds

$$\limsup_{\sigma\downarrow 0} \mathbb{E}_{w\sim\Pi_{\sigma,N,S}} R(\operatorname{clip} z_w) \quad \text{and} \quad \limsup_{\sigma\downarrow 0} R\left(\mathbb{E}_{w\sim\Pi_{\sigma,N,S}} \operatorname{clip} z_w\right).$$

The event is common to all widths. The limit in σ is pointwise in the learner width, with no width-uniform convergence rate asserted.

Proof. The zero network has empirical risk at most one. Consequently every global minimizer has energy $q = Q_N(w) \leq 1/\lambda$. Apply Lemma 2.2 at thresholds $K_j = 2^j$, $j = 0, \ldots, J$, each with failure probability $\delta/[2(J+1)]$. On the resulting event, common to every width, networks with $q \leq 1$ have deviation at most $e_n + b_n$. For $q > 1$, the least dyadic threshold containing q is less than $2q$, so their deviation is at most $a_n q^{H/2} + b_n$. In particular, for every $q \leq 1/\lambda$,

$$R(\operatorname{clip} z_w) \leq \widehat{R}_S(\operatorname{clip} z_w) + a_n q^{H/2} + e_n + b_n.$$

Since $H/2 \geq 1$ and $\lambda^{H/2} = a_n$,

$$a_n q^{H/2} \leq a_n \lambda^{1-H/2} q = \lambda q.$$

Clipping cannot increase squared loss against labels in $[-1, 1]$. Comparison of the minimizing objective with the cloned teacher gives

$$R(\operatorname{clip} z_w) \leq \widehat{R}_S(z_w) + \lambda Q_N(w) + e_n + b_n \leq \widehat{R}_S(g) + \lambda E + e_n + b_n.$$

Hoeffding's inequality gives (14) with failure probability at most $\delta/2$. For (15), put $Z = (g(X) - Y)^2$. Then $0 \leq Z \leq (U+1)^2$ and $\operatorname{Var}(Z) \leq (U+1)^2 R(g)$; Bernstein's inequality gives the stated bound. Union-bounding with the capacity event proves (13).

For completeness, at fixed N, S the function $F(w) = Q_N(w) + \lambda^{-1}\widehat{R}_S(z_w)$ is continuous and coercive, so its minimizer set is nonempty and compact. Write $f_* = \min F$. For each $\varepsilon > 0$, a

positive-volume neighborhood of a minimizer has $F \le f_* + \varepsilon/2$. If $\beta = (2\sigma^2)^{-1}$ and $\beta > \beta_0 > 0$, then

$$\int_{\{F \ge f_* + \varepsilon\}} e^{-\beta F} dw \le e^{-(\beta-\beta_0)(f_*+\varepsilon)} \int e^{-\beta_0 F} dw.$$

The final integral is finite because $F \ge Q_N$. Dividing by the normalizer lower bound from that neighborhood shows that the posterior mass of $\{F \ge f_* + \varepsilon\}$ tends to zero. Coercivity and continuity then show that the posterior expectation of any bounded continuous function has upper limit at most its maximum on the minimizer set. Apply this to $R(\operatorname{clip} z_w) \in [0, 4]$; its continuity follows by dominated convergence. Jensen's inequality gives the assertion for the mean of clipped outputs. □

The leading teacher-dependent term is explicitly

$$\lambda E = \left[\frac{2d(d+1)B^2}{H} + 4m^2B^2\right] \left(\frac{8D_n}{\sqrt{n}}\right)^{2/H}. \tag{16}$$

Thus, at fixed depth and for a fixed teacher, the excess risk is $\widetilde{O}(n^{-1/H}) = \widetilde{O}(n^{-1/(L-1)})$. For two hidden layers it is $\widetilde{O}(n^{-1/2})$. The likelihood variance is $n\lambda\sigma^2$ with the exact λ in (11); it is not a sample-independent constant. In particular, up to logarithms this calibration lies at the power $\lambda \asymp n^{-1/H}$.

Theorem 2.4 (One hidden layer)**.** *Suppose $H = 1$, and put*

$$a_n = 8\sqrt{2}\, D_n/\sqrt{n}, \qquad e_n = 8D_n/\sqrt{n}.$$

For any $\lambda > 0$, define J, b_n by (12). With the same assumptions and posterior interpretation as Theorem 2.3, the bound is

$$R(\operatorname{clip} z_w) \le R(g) + \lambda E + \frac{a_n^2}{4\lambda} + e_n + b_n + h_g. \tag{17}$$

In particular, $\lambda = a_n$ gives an excess of order $(E+1)D_n/\sqrt{n}$ plus the concentration terms. If a deterministic energy cap $E_0 \ge E$ is known, choosing

$$\lambda = \frac{a_n}{2\sqrt{E_0+1}} \tag{18}$$

replaces the two energy terms in (17) by at most $a_n\sqrt{E_0+1}$.

Proof. The same dyadic argument gives deviation at most $a_n\sqrt{q} + e_n + b_n$. After comparing penalized empirical risks, use

$$\sup_{q \ge 0}\{a_n\sqrt{q} - \lambda q\} = \frac{a_n^2}{4\lambda}.$$

The teacher concentration and posterior-limit arguments are unchanged. For the stated calibration, $\lambda E + a_n^2/(4\lambda) = a_n[E/(2\sqrt{E_0+1}) + \sqrt{E_0+1}/2] \le a_n\sqrt{E_0+1}$. □

2.3 Comparison with an expanding class of teachers

Corollary 2.5 (Growing teacher width and weight bound)**.** *Fix H, d and $0 < \delta < 1$. Let $\mathcal{G}_n$ be the class of networks with H hidden layers, width at most $\lfloor n^\alpha \rfloor$, and every weight bounded in absolute value by n^α.*

If $H \geq 2$, use (11). For every $0 \leq \alpha < 1/(4H)$, with probability at least $1-\delta$, simultaneously for all learner widths $N \geq \lfloor n^{\alpha} \rfloor$,

$$\limsup_{\sigma \downarrow 0} \mathbb{E}_{\Pi_{\sigma,N,S}} R(\operatorname{clip} z_w) \leq \min_{g \in \mathcal{G}_n} R(g) + \widetilde{O}_{H,d,\delta}(n^{-(1/H-4\alpha)}). \tag{19}$$

If $H = 1$, use (18) with $E_0 = [d(d+1)+2]n^{4\alpha}$. For every $0 \leq \alpha < 1/4$, the corresponding bound is

$$\limsup_{\sigma \downarrow 0} \mathbb{E}_{\Pi_{\sigma,N,S}} R(\operatorname{clip} z_w) \leq \min_{g \in \mathcal{G}_n} R(g) + \widetilde{O}_{d,\delta}(n^{-(1/2-2\alpha)}). \tag{20}$$

Both assertions also hold for the risk of the posterior mean of clipped outputs. The notation $\widetilde{O}$ suppresses logarithms in n; the preceding theorems give explicit finite-sample constants.

Proof. Narrower teachers embed by zero padding, since $\tanh(0) = 0$. The parameter set defining $\mathcal{G}_n$ is compact and its population risk is continuous, so it has a deterministic population-risk minimizer g_n. The zero network is an available comparator, hence $R(g_n) \leq 1$. Lemma 2.1 gives

$$E \leq [d(d+1)+2H]n^{4\alpha}, \qquad U \leq n^{2\alpha}.$$

The Bernstein term (15) is therefore $O_{\delta}(n^{2\alpha-1/2}+n^{4\alpha-1})$. For $H \geq 2$, (16) has order $\widetilde{O}_{H,d}(n^{4\alpha-1/H})$, which dominates the preceding concentration powers. For $H = 1$, the energy-calibrated bound in Theorem 2.4 has order $\widetilde{O}_{d}(n^{2\alpha-1/2})$. The dyadic concentration terms are at most logarithmic factors times $n^{-1/2}$ in both cases. □

What the exponent does and does not establish. The argument is independent of learner width and allows arbitrary label noise and arbitrary approximation error. It still uses a weight bound, or another bound on the teacher's energy, as well as its width. It does not give a bound in teacher width alone for arbitrary real-valued networks. It also does not make the dependence on depth polynomial: the leading sample requirement from (16) raises the teacher-energy and accuracy factors to the power H.

The distinction between clipping before and after posterior averaging is substantive. The bounded-loss argument above controls $\mathbb{E}_{\Pi} R(\operatorname{clip} z_w)$ and $R(\mathbb{E}_{\Pi} \operatorname{clip} z_w)$. By itself it does not control $R(\mathbb{E}_{\Pi} z_w)$ or $R(\operatorname{clip}(\mathbb{E}_{\Pi} z_w))$.

3 A finite-scale Gaussian result with no prior-strength limit

A related oracle inequality holds for an actual finite-scale posterior, without sending a prior parameter to zero. The proof below applies to all learner widths at once when the prior variances decay sufficiently quickly with width. It controls clipped posterior draws and the mean of clipped outputs. Its likelihood still uses the raw real-valued network output.

Use the network architecture, energy Q_N, and risks from Section 2. In this section inputs lie in $[-1,1]^d$ and labels lie in $[-1,1]$; their joint law is otherwise arbitrary. For $\gamma \geq 2$—and also for $\gamma \geq 1$ in the one-hidden-layer result below—let the prior $\mu_{N,\gamma}$ have independent centered Gaussian coordinates with variances

$$\operatorname{Var}(W_{jk}^{1}) = \frac{N^{1-\gamma}}{d}, \qquad \operatorname{Var}(v_j) = N^{-\gamma-1},$$
$$\operatorname{Var}(W_{jk}^{\ell}) = N^{-\gamma} \quad (2 \leq \ell \leq H). \tag{21}$$

Thus this is the Gaussian family of (2) with $\sigma^2 = N^{-\gamma}$. For $\lambda > 0$, use a Gaussian likelihood with variance $n\lambda N^{-\gamma}$ and mean $z_w(X_i)$. The posterior is

$$\Pi_{N,\gamma,S}(dw) = \frac{\exp[-\beta_N \widehat{R}_S(z_w)]}{Z_{N,\gamma,S}} \mu_{N,\gamma}(dw), \qquad \beta_N = \frac{N^\gamma}{2\lambda}. \tag{22}$$

All statements below concern this probability measure at its specified finite width and scale.

3.1 An explicit finite-scale oracle inequality

Let g be a teacher of hidden width m, with weights bounded by $B \geq 1$. Define

$$\begin{aligned} E &= d(d+1)B^2 + 2Hm^2B^2, \qquad U = mB, \\ D_0 &= d + H + 1, \qquad A_g = \sqrt{2}mB, \\ K_g &= 1 + A_g \left(\sqrt{2} A_g^{H-1} + \sum_{j=0}^{H-2} A_g^j \right). \end{aligned} \tag{23}$$

The sum is empty if $H = 1$. In particular, $K_g \geq 1$ and $\log K_g = O(H \log(2 + mB))$. For $H \geq 2$, put

$$\begin{aligned} p &= H/2, \qquad A_n = \frac{8D_n}{H^p\sqrt{n}}, \qquad J_\lambda = \left\lceil \log_2 \frac{8}{\lambda} \right\rceil, \\ t_\lambda &= \log \frac{2(J_\lambda + 1)}{\delta}, \qquad r_{n,\lambda} = A_n + 4\sqrt{\frac{t_\lambda}{2n}}. \end{aligned} \tag{24}$$

For the scalar teacher concentration term, write

$$b_n(g,\delta) = (U+1)\sqrt{\frac{2R(g)\log(2/\delta)}{n}} + \frac{2(U+1)^2 \log(2/\delta)}{3n}. \tag{25}$$

Theorem 3.1 (A finite-scale oracle inequality, uniform over widths)**.** *Let $H \geq 2$, $n \geq 2$, $0 < \delta < 1$, and suppose*

$$\frac{1}{n} \leq \lambda \leq 1, \qquad \lambda^p \geq 2^{4p-2} A_n. \tag{26}$$

For every fixed teacher g, with probability at least $1 - \delta$, simultaneously for every $N \geq m$ and every $\gamma \geq 2$,

$$\begin{aligned} \int R(\operatorname{clip} z_w)\, \Pi_{N,\gamma,S}(dw) \leq R(g) + b_n(g,\delta) + r_{n,\lambda} \\ + \lambda \left[E + 1 + D_0 \log 2 + 2D_0 \log \frac{4K_g(U+1)}{\lambda} \right]. \end{aligned} \tag{27}$$

The same right-hand side bounds the risk of the posterior mean of clipped outputs,

$$x \longmapsto \int \operatorname{clip}(z_w(x))\, \Pi_{N,\gamma,S}(dw).$$

For example, take

$$\lambda_n = \max\left\{ \frac{1}{n}, \left(2^{4p-2} A_n\right)^{1/p} \right\}, \tag{28}$$

for sufficiently large n that $\lambda_n \leq 1$. At fixed d, H, δ and teacher, the excess in (27) is $\widetilde{O}(n^{-1/H})$. The bound has no learner-width factor and requires no limit in either N or the prior scale.

3.2 A deviation inequality valid at every energy

The first step strengthens the deviation bound used for the strong-prior minimizers: here it must hold over the full parameter space because the finite-scale posterior has unbounded support.

Lemma 3.2 (Global clipped deviation)**.** *Under* (26)*, with probability at least* $1-\delta/2$*, simultaneously for every network at every width,*

$$R(\operatorname{clip} z_w) \le \widehat{R}_S(z_w) + \frac{\lambda}{2} Q_N(w) + r_{n,\lambda}. \tag{29}$$

Proof. Apply (8) to the deterministic classes $\mathcal{Z}_{2^j}$, $0 \le j \le J_\lambda$, with individual failure probability $\delta/[2(J_\lambda+1)]$. On the intersection event, write $k = Q_N(w)$. For $k \le 1$, the unit-energy class gives the claim, since clipping cannot increase empirical square loss. For $1 < k \le 8/\lambda$, the appropriate shell has radius at most $2k$, so its capacity term is at most

$$A_n 2^p k^p \le A_n 2^p (8/\lambda)^{p-1} k \le \frac{\lambda}{2} k,$$

where the last inequality is exactly $\lambda^p \ge 2^{4p-2} A_n$. Finally, for $k > 8/\lambda$, the clipped population loss is at most four and the empirical loss is nonnegative; therefore

$$R(\operatorname{clip} z_w) - \widehat{R}_S(z_w) \le 4 < \frac{\lambda}{2} k.$$

This also covers arbitrarily large energies. □

3.3 The posterior's variational comparison

There are

$$P_N = N(d+2) + (H-1)N^2$$

parameters. Under the Gaussian prior, $N^\gamma Q_N$ has a chi-square distribution with P_N degrees of freedom. Thus

$$\log \int \exp\left(\frac{N^\gamma Q_N}{4}\right) d\mu_{N,\gamma} = \frac{P_N}{2} \log 2.$$

The entropy variational inequality implies, for every comparison measure Q,

$$\mathrm{KL}(Q \| \mu_{N,\gamma}) \ge \frac{N^\gamma}{4} \int Q_N \, dQ - \frac{P_N}{2} \log 2. \tag{30}$$

The posterior (22) minimizes

$$Q \longmapsto \int \widehat{R}_S(z_w)\, dQ + 2\lambda N^{-\gamma}\, \mathrm{KL}(Q \| \mu_{N,\gamma}).$$

Combining its variational optimality with (30) gives, for every comparison measure Q of finite relative entropy,

$$\begin{aligned} &\int \left[\widehat{R}_S(z_w) + \frac{\lambda}{2} Q_N(w)\right] \Pi_{N,\gamma,S}(dw) \\ &\qquad \le \int \widehat{R}_S(z_w)\, Q(dw) + 2\lambda N^{-\gamma}\, \mathrm{KL}(Q \| \mu_{N,\gamma}) + \lambda \frac{P_N}{N^\gamma} \log 2. \end{aligned} \tag{31}$$

All quantities are finite for the Gaussian comparison measure used below. They are also finite for the posterior: its density relative to the prior is bounded, and the prior has finite energy moments. For $\gamma \geq 2$ and $N \geq 1$,

$$\frac{P_N}{N^\gamma} \leq d + 2 + H - 1 = D_0. \tag{32}$$

For $H \geq 2$, this is the point at which the condition $\gamma \geq 2$ enters. When $H = 1$, the quadratic term in P_N is absent: $P_N/N^\gamma \leq d + 2 = D_0$ already holds for every $\gamma \geq 1$.

3.4 A Gaussian comparison measure near the teacher

Let w_* be the exact teacher clone from Lemma 2.1. Denote the prior covariance by Σ_N. For $0 < r \leq 1$, define

$$Q_r = \mathcal{N}(w_*, r^2 \Sigma_N).$$

The elementary Gaussian relative-entropy formula gives

$$\mathrm{KL}(Q_r \| \mu_{N,\gamma}) = \frac{N^\gamma Q_N(w_*)}{2} + \frac{P_N}{2}(r^2 - 1 - \log r^2) \leq \frac{N^\gamma E}{2} + P_N \log(1/r). \tag{33}$$

We now bound the comparison measure's prediction error. The clone's internal matrices have operator norm at most $A_g = \sqrt{2}mB$, and $\sqrt{N}\|v_*\|_2 \leq A_g$. For the internal matrices, identify each group of clones with its normalized indicator vector. The induced matrix has entries $T_{ji}\sqrt{k_j/k_i}$, whose Frobenius norm is at most $\sqrt{2}\,\|T\|_F \leq \sqrt{2}mB$; the readout bound follows directly from the cloning construction. For fixed $x \in [-1, 1]^d$, put

$$e_\ell(x) = \left(\frac{1}{N} \mathbb{E}_{Q_r} \|h^\ell(x) - h_*^\ell(x)\|_2^2 \right)^{1/2}.$$

The activation is 1-Lipschitz. In the first layer,

$$e_1(x) \leq r N^{(1-\gamma)/2} \frac{\|(1, x)\|_2}{\sqrt{d}} \leq \sqrt{2}\, r. \tag{34}$$

For later hidden layers, use

$$W^\ell h^{\ell-1} - W_*^\ell h_*^{\ell-1} = W_*^\ell (h^{\ell-1} - h_*^{\ell-1}) + (W^\ell - W_*^\ell) h^{\ell-1}.$$

The noise matrix in the second term is independent of $h^{\ell-1}$, and $\|h^{\ell-1}\|_2^2 \leq N$. Hence Minkowski's inequality and its Gaussian second moment give

$$e_\ell(x) \leq A_g e_{\ell-1}(x) + r N^{(1-\gamma)/2} \leq A_g e_{\ell-1}(x) + r. \tag{35}$$

Readout noise is likewise independent of the last hidden layer and has prediction variance at most $r^2 N^{-\gamma}$. Consequently

$$\begin{aligned} \left(\mathbb{E}_{Q_r} |z_w(x) - g(x)|^2 \right)^{1/2} &\leq A_g e_H(x) + r N^{-\gamma/2} \\ &\leq r \left[1 + A_g \left(\sqrt{2} A_g^{H-1} + \sum_{j=0}^{H-2} A_g^j \right) \right] = r K_g. \end{aligned} \tag{36}$$

This is a uniform *pointwise second-moment* bound. It does not assert that one draw is uniformly close to the teacher over all inputs; that stronger assertion is unnecessary here.

Choose

$$\rho = \frac{\lambda}{4(U+1)}, \qquad r = \frac{\rho}{K_g}. \tag{37}$$

These satisfy $0 < r \le 1$. Minkowski's inequality in the product of Q_r and the empirical input measure gives

$$\begin{aligned}\int \widehat{R}_S(z_w)\, Q_r(dw) &\le \widehat{R}_S(g) + 2\rho\sqrt{\widehat{R}_S(g)} + \rho^2 \\ &\le \widehat{R}_S(g) + 2\rho(U+1) + \rho^2 \le \widehat{R}_S(g) + \lambda. \end{aligned} \tag{38}$$

Proof of Theorem 3.1. Integrate the global deviation inequality (29) against the posterior, and then apply (31) with $Q = Q_r$. Equations (32), (33), and (38) imply

$$\begin{aligned}\int R(\operatorname{clip} z_w)\, \Pi_{N,\gamma,S}(dw) &\le \widehat{R}_S(g) + r_{n,\lambda} \\ &\quad + \lambda\left[E + 1 + D_0 \log 2 + 2D_0 \log \frac{4K_g(U+1)}{\lambda}\right].\end{aligned}$$

For the fixed teacher, Bernstein's inequality gives $\widehat{R}_S(g) \le R(g) + b_n(g,\delta)$ except on an event of probability $\delta/2$. Intersecting this event with the global deviation event proves (27). Neither event depends on N or γ, so the conclusion is simultaneous over all the claimed widths and scales. Jensen's inequality proves the claim for the mean of clipped outputs. The rate follows from (28); $\log(1/\lambda_n)$ adds a logarithm, without changing its power of n. □

3.5 One hidden layer and the limits of this argument

At $H = 1$, the finite-scale result holds for *every* $\gamma \ge 1$, including the mean-field scale $\gamma = 1$, with rate $\widetilde{O}(n^{-1/2})$. Its guarantee is simultaneous over all $N \ge m$ and all these γ. The wider range follows because $P_N = (d+2)N$, so $P_N/N^\gamma \le d+2$ for $\gamma \ge 1$. The Gaussian perturbation estimates also hold throughout this range: $N^{(1-\gamma)/2} \le 1$ and $N^{-\gamma/2} \le 1$ are their only scale-dependent requirements. In (24) take $p = 1/2$ and replace the residual by

$$r_{n,\lambda} = A_n + \frac{A_n^2}{\lambda} + 4\sqrt{\frac{t_\lambda}{2n}}.$$

There is no condition $\lambda^p \ge 2^{4p-2}A_n$ in this case. The needed shell estimate is simply

$$A_n\sqrt{2k} \le \frac{\lambda}{2}k + \frac{A_n^2}{\lambda}.$$

The unit-energy and high-energy cases are unchanged, and all subsequent steps of the proof hold verbatim. Choosing $\lambda_n = \max\{A_n, 1/n\}$, for sufficiently large n that $\lambda_n \le 1$, gives the stated rate for a fixed teacher.

The finite-scale theorem proves a statement about clipped draws and means of clipped outputs. It makes no claim about the raw posterior mean at finite scale. For two or more hidden layers, this proof does not cover the entire range $\gamma \ge 1$: when $\gamma < 2$, the quadratic parameter-count term makes P_N/N^γ grow with the learner width. Removing that term requires additional control in function space. With one hidden layer, that quadratic term is absent, and the theorem already covers every $\gamma \ge 1$. The raw-output results of Section A instead concern the strong-prior limit at each fixed width and therefore do not need $\gamma \ge 2$.

4 A sharp oracle bound for the final Bayesian posterior mean

A different choice of prior and likelihood gives a substantially faster approximation guarantee. The predictor in this section is ordinary Bayesian model averaging after all observations have been supplied. It uses neither a randomly selected training prefix nor an average of the posteriors at successive times. The guarantee is in expectation over the training sample.

The underlying exponential-weights result is closely related to the recent theorem of Høgsgaard, Rebeschini and Wegel [2], who prove optimal expected aggregation rates for a finite dictionary. We give a self-contained arbitrary-prior version, with a conservative temperature constant, so that it applies directly to continuous neural-network parameter distributions. The point of the application below is the explicit neural-network prior and its dependence on teacher width, depth and weight magnitude.

Let π be a probability measure on a measurable parameter space Θ, and let $(h, x) \mapsto f_h(x)$ be jointly measurable with values in $[-1, 1]$. Let D be any probability law of (X, Y) with $Y \in [-1, 1]$. For any bounded measurable predictor f, write $R(f) = \mathbb{E}_D(f(X) - Y)^2$, and abbreviate $R(h) = R(f_h)$. Define

$$R(h) = \mathbb{E}_D(f_h(X) - Y)^2, \qquad \widehat{R}_S(h) = \frac{1}{n}\sum_{i=1}^{n}(f_h(X_i) - Y_i)^2.$$

For n independent observations $S \sim D^n$, define

$$p_S(dh) = \frac{\exp\{-\eta n \widehat{R}_S(h)\}\pi(dh)}{\int \exp\{-\eta n \widehat{R}_S(u)\}\pi(du)}, \qquad \widehat{f}_S(x) = \int f_h(x) p_S(dh). \tag{39}$$

For $n = 0$, the posterior is understood to be π.

Theorem 4.1 (Final posterior mean; arbitrary prior)**.** *Suppose $0 < \eta \le \frac{1}{4}\log(5/4)$. For every integer $n \ge 0$,*

$$\mathbb{E}_{S\sim D^n} R(\widehat{f}_S) \le \inf_Q \left\{ \int R(h) Q(dh) + \frac{\mathrm{KL}(Q\|\pi)}{\eta(n+1)} \right\}. \tag{40}$$

The infimum is over probability measures Q independent of S; they may depend on D. In particular, $\eta = 1/20$ is admissible.

The leading coefficient of the comparison risk in (40) is one. The theorem allows arbitrary misspecification and arbitrary dependence of the label on the input. It concerns the posterior *mean*; the same conclusion is not asserted for a random posterior draw. It is also an expectation bound, not a high-probability bound.

Lemma 4.2 (Deleting one observation)**.** *Let q be any probability law on $[-1, 1]$, let $y \in [-1, 1]$, and set*

$$p(da) = \frac{e^{-\eta(a-y)^2} q(da)}{\int e^{-\eta(u-y)^2} q(du)}.$$

If $4(e^{4\eta} - 1) \le 1$, then

$$\left(\int a\, q(da) - y \right)^2 \le \int (a - y)^2 p(da). \tag{41}$$

Proof. Write $\ell(a) = (a - y)^2 \in [0, 4]$, and interpolate from p back to q by

$$q_t(da) = \frac{e^{t\eta\ell(a)} p(da)}{\int e^{t\eta\ell(u)} p(du)}, \qquad 0 \le t \le 1.$$

Thus $q_0 = p$, $q_1 = q$, and boundedness permits differentiation under the integral:

$$\int \ell \, dq - \int \ell \, dp = \eta \int_0^1 \mathrm{Var}_{q_t}(\ell) \, dt.$$

The function $a \mapsto (a-y)^2$ is 4-Lipschitz on $[-1,1]$. The independent-pair formula for variance gives $\mathrm{Var}_{q_t}(\ell) \le 16 \, \mathrm{Var}_{q_t}(a)$. Furthermore,

$$\frac{dq_t}{dq} \le e^{4\eta(1-t)}.$$

If probability measures ν, μ satisfy $\nu \le K\mu$, then $\mathrm{Var}_\nu(a) \le K \, \mathrm{Var}_\mu(a)$: use $\mathrm{Var}_\nu(a) = \inf_b \int (a-b)^2 d\nu$ and choose $b = \int a \, d\mu$. Consequently,

$$\int \ell \, dq - \int \ell \, dp \le 16\eta \int_0^1 e^{4\eta(1-t)} dt \; \mathrm{Var}_q(a) = 4(e^{4\eta} - 1) \, \mathrm{Var}_q(a).$$

Finally, $(\int a \, dq - y)^2 = \int \ell \, dq - \mathrm{Var}_q(a)$. The assumed bound on η proves the claim. □

Proof of Theorem 4.1. Use $n+1$ independent observations, with full posterior p and posterior q_i obtained by deleting observation i. Conditional on all these observations, the image of p under $h \mapsto f_h(X_i)$ is the exponential tilt in Lemma 4.2 of the corresponding image of q_i. Hence, pointwise in the sample,

$$\frac{1}{n+1} \sum_{i=1}^{n+1} \left(\int f_h(X_i) q_i(dh) - Y_i \right)^2 \le \int \widehat{R}_{S_{n+1}}(h) p(dh).$$

The Gibbs variational principle gives, for every fixed comparison measure Q with finite relative entropy,

$$\int \widehat{R}_{S_{n+1}} \, dp \le \int \widehat{R}_{S_{n+1}} \, dQ + \frac{\mathrm{KL}(Q\|\pi)}{\eta(n+1)}.$$

On taking expectations, each summand on the first display becomes $\mathbb{E}_{S\sim D^n} R(\widehat{f}_S)$: q_i uses n independent observations and is independent of observation i. The fixed comparison measure satisfies $\mathbb{E} \int \widehat{R}_{S_{n+1}} dQ = \int R \, dQ$. This proves the bound for every Q, and therefore for its infimum. For the convenient constant, $e^{1/5} \le (1 - 1/5)^{-1} = 5/4$, so $\eta = 1/20$ satisfies the hypothesis. □

5 A continuous architecture prior with teacher-dependent rates

We now specify a Bayesian neural-network learner for which the oracle penalty depends on the teacher's parameters, not the largest architecture available to the learner. This is a prior on network architecture and weights. It differs from a fixed-width prior that independently draws all weights from one Gaussian law.

5.1 Network class and prior

Inputs lie in any Borel subset $\mathcal{X} \subseteq [-1,1]^d$, where $d \ge 1$. Fix a globally Lipschitz activation φ, and choose $a \ge \max\{1, \mathrm{Lip}(\varphi)\}$ and $b = |\varphi(0)|$. No condition $\varphi(0) = 0$ is needed. A depth-L, width-m network, with $L \ge 2$ and $m \ge 1$, has $L-1$ hidden layers of width m and a linear readout:

$$h^0(x) = (1, x), \qquad h^\ell(x) = \varphi(W^\ell h^{\ell-1}(x)), \quad 1 \le \ell < L, \qquad z_w(x) = v^\mathsf{T} h^{L-1}(x).$$

The activation is applied coordinatewise. The input augmentation provides first-layer biases; there are no other biases. The parameter count is

$$P_{L,m} = m(d+1) + (L-2)m^2 + m. \tag{42}$$

Its bounded prediction is $f_w(x) = \mathrm{clip}(z_w(x))$, where $\mathrm{clip}(u) = \max\{-1, \min\{1, u\}\}$.

Let $N \in \mathbb{N} \cup \{\infty\}$ be the maximum allowed width, and $L_{\max} \in \{2, 3, \ldots\} \cup \{\infty\}$ the maximum allowed depth. On allowed pairs (L, m) put prior mass proportional to

$$\omega_{L,m} = \frac{1}{L(L-1)m(m+1)}, \qquad 2 \le L \le L_{\max}, \quad 1 \le m \le N. \tag{43}$$

These weights sum to one when $N = L_{\max} = \infty$. Conditioning on finite architecture limits only increases the probability of each allowed pair. Given (L, m), draw its $P_{L,m}$ weights independently from the standard Cauchy law, with density $u \mapsto 1/[\pi(1+u^2)]$. Call the resulting proper prior $\pi_{N,L_{\max}}$. The parameter space is the disjoint union of the allowed architecture spaces. In particular, varying L selects networks of genuinely different depths; it does not require embedding a shallower tanh network into one fixed depth-$L_{\max}$ architecture.

The learner uses the likelihood

$$\prod_{i=1}^{n} \exp\left\{-\frac{1}{20}(f_w(X_i) - Y_i)^2\right\} \tag{44}$$

and predicts with the final posterior mean of f_w. This is the usual Gaussian likelihood for a clipped network mean and observation variance 10, up to factors independent of w. The theorem below imposes no Gaussian assumption on the actual labels. In particular it permits arbitrary label noise. The learner does not require exact interpolation of the observed labels.

Theorem 5.1 (Agnostic neural-network oracle bound)**.** *Let D be any probability law on $\mathcal{X} \times [-1, 1]$, and let $\widehat{f}_{n,N,L_{\max}}$ be the preceding posterior-mean predictor. Let $g = z_{w_*}$ be any network of depth $2 \le L \le L_{\max}$ and width $1 \le m \le N$, all of whose weights have absolute value at most $B \ge 1$. Put*

$$D_{d,m} = \max\{d+1, m\}, \qquad A = 1 + b + aD_{d,m}(B+1), \qquad K = LA^L,$$

and let $P = P_{L,m}$ as in (42). For every $0 < \rho \le 1$ and $n \ge 0$,

$$\mathbb{E}R(\widehat{f}_{n,N,L_{\max}}) \le R(\mathrm{clip}\, g) + 4\rho \tag{45}$$

$$+ \frac{20}{n+1}\left[\log(L(L-1)m(m+1)) + P\log\left(\frac{\pi[1+(B+1)^2]K}{2\rho}\right)\right]. \tag{46}$$

The same inequality holds with $R(g)$ in place of $R(\mathrm{clip}\, g)$. Its right side is independent of N and $L_{\max}$ apart from the requirement that they admit the competitor.

The theorem holds for every choice of architecture limits, including no finite limits at all. Because it is an expectation bound, it does not assert a common high-probability sample event for all choices of these limits. Taking the infimum in (46) permits a different competitor at every sample size.

Lemma 5.2 (Parameter perturbations)**.** *In the setting of Theorem 5.1, if $\|w - w_*\|_\infty \le r \le 1$, then*

$$\sup_{x \in \mathcal{X}} |z_w(x) - z_{w_*}(x)| \le Kr.$$

Proof. Both parameter vectors have coordinates bounded by $B+1$. Let M_ℓ bound the maximum coordinate magnitude in hidden layer ℓ of either network, and let δ_ℓ bound their coordinatewise difference, uniformly in x. Since $M_0 \le 1$, the activation bound $|\varphi(u)| \le b + a|u|$ gives $M_\ell \le A^\ell$ by induction. The perturbation recurrence is

$$\delta_\ell \le aD_{d,m} r A^{\ell-1} + aD_{d,m}(B+1)\delta_{\ell-1}, \qquad \delta_0 = 0.$$

Both $aD_{d,m}$ and $aD_{d,m}(B+1)$ are at most A, so induction gives $\delta_\ell \le \ell r A^\ell$. For the readout, use $D_{d,m} \le A$ and $D_{d,m}(B+1) \le A$ to obtain

$$|z_w - z_{w_*}| \le D_{d,m} r A^{L-1} + D_{d,m}(B+1)(L-1) r A^{L-1} \le L r A^L.$$

□

Proof of Theorem 5.1. Choose a comparison measure supported on architecture (L,m), uniform on the parameter cube $\{w : \|w - w_*\|_\infty \le r\}$ with $r = \rho/K \le 1$. Lemma 5.2 and the fact that clipping is nonexpansive show $\|f_w - \operatorname{clip} g\|_\infty \le \rho$ throughout this cube. Squared loss is 4-Lipschitz in its prediction on $[-1,1]$, hence

$$\int R(f_w) Q(dw) \le R(\operatorname{clip} g) + 4\rho.$$

On the cube, each Cauchy coordinate density is at least $1/\{\pi[1+(B+1)^2]\}$. The uniform density there is $(2r)^{-P}$. Moreover, the architecture has prior mass at least $\omega_{L,m}$. Therefore

$$\mathrm{KL}(Q \| \pi_{N,L_{\max}}) \le \log \frac{1}{\omega_{L,m}} + P \log\left(\frac{\pi[1+(B+1)^2]}{2r}\right).$$

Apply Theorem 4.1 with $\eta = 1/20$ and substitute $r = \rho/K$. Finally, clipping cannot increase squared error against a label in $[-1,1]$, so $R(\operatorname{clip} g) \le R(g)$. □

5.2 The same rate with conditionally Gaussian weights

The logarithmic cost of the teacher's weight magnitude does not require Cauchy coordinates. There is also a Gaussian scale-mixture version of the prior. After selecting (L,m) with the same architecture law, choose an integer $j \ge 0$ with probability $1/[(j+1)(j+2)]$, and then draw the weights independently from $N(0, 4^j)$. The prior is proper, and its coordinates are independent conditional on the architecture and scale.

Corollary 5.3 (Gaussian scale mixture)**.** *Use this Gaussian scale-mixture prior with the likelihood and predictor of (44). For the competitor in Theorem 5.1, let $j_B = \lceil \log_2 B \rceil$. Then*

$$\mathbb{E} R(\widehat{f}_{n,N,L_{\max}}) \le R(\operatorname{clip} g) + 4\rho \tag{47}$$

$$+ \frac{20}{n+1}\left[\log(1/\omega_{L,m}) + \log((j_B+1)(j_B+2)) + P \log\left(\frac{B\sqrt{2\pi}e^2 K}{\rho}\right)\right]. \tag{48}$$

Proof. The comparison measure selects architecture (L,m), selects scale j_B, and uses the same uniform parameter cube of radius $r = \rho/K$. Put $s = 2^{j_B}$, so $B \le s \le 2B$. Every coordinate in the cube satisfies $|w_i| \le B+1 \le 2s$; its $N(0,s^2)$ density is therefore at least $e^{-2}/(s\sqrt{2\pi})$. The coordinate contribution to relative entropy is at most

$$P \log\left(\frac{s\sqrt{2\pi}e^2}{2r}\right) \le P \log\left(\frac{B\sqrt{2\pi}e^2}{r}\right).$$

Adding the architecture and scale selection costs gives the bracket in (48). The risk comparison and final posterior-mean theorem apply as before. □

The additional scale cost is only $O(1+\log(1+\log B))$. Both prior choices consequently have the same polynomial width–depth rates described next. A single fixed Gaussian scale would instead charge order PB^2 for translating the comparison cube; the scale mixture makes this cost logarithmic in B.

5.3 Rates, activations, and what changes in the learner

Setting $\rho = 1/(n+1)$ in (46) yields a comparison penalty of order

$$\frac{1}{n+1}\left\{1+\log(Lm)+P_{L,m}\left[\log(n+1)+\log(B+1)+\log L+L\log A\right]\right\}. \tag{49}$$

Here $A = 1+b+a\max\{d+1,m\}(B+1)$, as above, and the hidden constant is numerical. Thus the cost is approximately one bit per parameter per required precision bit, together with the architecture description. The dependence on the weight bound is logarithmic. No bound on B must be selected in advance by the learner.

For fixed activation, input dimension and depth, competitors of width $m \le n^{\alpha}$ and weights bounded by n^{β}, where β is any fixed nonnegative constant, have excess expected risk $O(n^{2\alpha-1}\log n)$. Every $\alpha < 1/2$ is therefore admissible. For a single hidden layer ($L=2$), $P_{2,m} = m(d+2)$, and this improves to $O(n^{\alpha-1}\log n)$: every $\alpha < 1$ is admissible. If the depth also varies, the displayed bound has polynomial, rather than exponential, depth dependence. Suppressing logarithms and holding d, B and the activation fixed, it is $\widetilde{O}((mdL+m^2L^2)/(n+1))$. These statements are approximation guarantees: the target distribution need not be representable by any network.

The activation assumption includes tanh and ReLU with $a=1, b=0$; GELU, $\varphi(u) = u\Phi(u)$, with $a=2, b=0$; and softplus with $a=1, b=\log 2$. In particular, the proof neither relies on bounded hidden activations nor requires $\varphi(0)=0$. The Gaussian derivative bound $|\Phi(u)+u\phi(u)| \le 1+(2\pi e)^{-1/2} < 2$ verifies the stated GELU constant. Internal biases can be added by augmenting each layer with a constant coordinate and changing the parameter count and $D_{d,m}$ accordingly.

The price of this improved bound is a change of prior and prediction convention, not an unproved property of the original dense Gaussian prior. A draw from $\pi_{N,L_{\max}}$ first selects the *active* architecture and then draws its weights. At a fixed depth and maximum width it can equivalently be regarded as a structural sparsity prior, with exact zero connections outside the selected subnetwork. It is not a product law on all the weights of a dense width-N network. Its preference for small architectures is explicit. The guarantee therefore does not resolve whether an unrestricted dense Gaussian prior has a comparable polynomial-depth approximation theorem.

Finally, the loss in the likelihood is computed from the clipped network output, and the predictor averages those clipped outputs. Replacing either with an unrestricted real output is a different learner. Boundedness is used in Lemma 4.2; no such replacement follows from that lemma.

6 What these guarantees establish

6.1 Approximation rather than exact realization

The comparison network need not describe the entire data-generating process. It may simply be a useful approximation. For example, at fixed depth and dimension, the architectural-prior theorem gives

$$\mathbb{E}R(\widehat{f}_n) \le \inf_g \left\{ R(\operatorname{clip} g) + \widetilde{O}\left(\frac{m(g)^2}{n}\right)\right\},$$

when the comparison weights grow at most polynomially with n; the logarithms and depth factors are specified in Theorem 5.1. Increasing the teacher width can reduce approximation error while increasing the estimation penalty. This is the usual approximation–estimation tradeoff, now stated for a particular Bayesian learner. With a single hidden layer the number of parameters is linear in width, so the corresponding penalty is linear in m at fixed dimension.

The Gaussian theorem obtains the same kind of comparison without introducing a prior over architectures, but with a slower depth-dependent rate. The two guarantees therefore answer different versions of the learning question. Neither supplies an efficient posterior-sampling algorithm or an optimization guarantee for gradient descent.

6.2 Why the likelihood is not an interpolation requirement

With arbitrary label noise, fitting every observation closely is a different objective from predicting its conditional mean. For the Gaussian learner, the likelihood variance is $n\lambda_n\sigma^2$. Although this tends to zero as $\sigma \downarrow 0$, the prior simultaneously strengthens; the posterior selects minimizers of empirical squared loss plus $\lambda_n Q_N$, rather than exact interpolants. The regularization term is essential to our proof.

For the architectural prior we instead use a fixed observation variance of 10, after normalizing labels and predictions to $[-1, 1]$. This is an ordinary Gaussian observation likelihood used under possible misspecification. Its relatively weak per-observation weighting permits the posterior-mean averaging argument. Neither construction assumes that the actual data noise is Gaussian or has the likelihood's variance.

Earlier counterexamples [1] show that, at depth at least three and sufficiently large learner width, the untempered strong-prior objective can memorize independent label noise and have population error bounded away from that of the zero predictor. The positive results here do not turn exact noisy interpolation into a generally valid agnostic learning rule.

6.3 A width-only comparison needs a qualification on weights

Our Gaussian bound charges teacher weight magnitude polynomially. The architectural-prior bound charges it only logarithmically. Some dependence on precision or weight magnitude is necessary for a distribution-uniform guarantee for a fixed proper Bayesian prior on continuous functions.

Proposition 6.1 (An obstruction from nearby inputs)**.** *Fix $n < \infty$, $\eta > 0$, and any proper prior π on continuous predictors $f : \mathbb{R} \to [-1, 1]$, with measurable evaluation. Use the posterior proportional to $\exp[-\eta \sum_{i=1}^{n}(f(X_i) - Y_i)^2]\pi(df)$. Let D_r assign probability $1/2$ to each of $(-r, -1)$ and $(r, 1)$. As $r \downarrow 0$, uniformly over every sample of size n from D_r, the posterior-mean risk and the posterior-averaged risk satisfy*

$$\liminf R_{D_r}(\widehat{f}_S) \geq 1, \qquad \liminf \int R_{D_r}(f)\, p_S(df) \geq 1.$$

Nevertheless a width-one tanh network at any prescribed fixed depth has zero risk under D_r, if its weights may grow as $r \downarrow 0$.

Proof. The likelihood lies in $[e^{-4\eta n}, 1]$, so $dp_S/d\pi \leq e^{4\eta n}$, uniformly over the sample. Continuity and bounded convergence give

$$a_r := \int |f(r) - f(-r)|\, \pi(df) \longrightarrow 0.$$

For every bounded predictor,

$$R_{D_r}(f) = 1 + \tfrac{1}{2}(f(-r)^2 + f(r)^2) + f(-r) - f(r) \geq 1 - |f(r) - f(-r)|.$$

Integrating proves the posterior-draw bound $1 - e^{4\eta n} a_r$. The posterior mean satisfies the same bound by the triangle inequality. For the teacher, set the first scalar weight to $1/r$, the constant-input weight to zero, and all remaining hidden weights to one. If T denotes tanh composed $L-1$ times, choose readout weight $1/T(1)$. Its two outputs are exactly -1 and 1. Its first weight grows like $1/r$. □

The obstruction applies to the architectural priors above as well: every sampled finite network is continuous, even when the prior has no maximum width or depth. It does not contradict their oracle bounds, whose penalty depends on the large teacher weight. It is not an impossibility result for every learning algorithm; it concerns the stated fixed-prior Bayesian rule. The same obstruction holds on the unit circle: use the inputs $(\pm r, \sqrt{1-r^2})$, which coalesce, and the same teacher applied to the first coordinate.

6.4 Expectation, high probability, and the remaining Gaussian question

The Gaussian result provides a high-probability bound on one sample event for all eligible learner widths. The architectural-prior result instead bounds the expected risk of each learner. These quantifiers cannot be interchanged. In particular, the latter proof does not give the same rate with only a $\log(1/\delta)$ confidence penalty, and its posterior-draw risk need not obey the same bound. The posterior mean can have lower risk than every individual member of its support; its improvement is used explicitly in the leave-one-out proof.

For dense independent Gaussian priors, the principal remaining question is whether one can obtain a comparable teacher-parameter-count oracle bound without explicitly selecting a small architecture. The present paper improves the strong-prior rate, but does not prove such a fast rate or a joint polynomial-depth guarantee for that learner. The architectural result is not a resolution of this Gaussian-prior question.

Verification and provenance. The positive theorem in the supplied tempered-learner note was checked against its complete supporting proof [1]; no error was found in that theorem under its stated conventions. The stronger results here have independent AI-agent mathematical reviews and reproducible source builds. They have not been formalized in Lean and have not undergone external peer review. The supporting audit records distinguish this scope from the collection's formally verified results.

A The same sample exponent without clipping

The Gaussian-prior model and likelihood family in Section 2, with the calibration below, also admit an oracle inequality for its *raw* real-valued predictions. This covers both posterior-averaged risk and the ordinary posterior mean $\int z_w \, d\Pi_{\sigma,N,S}$, with no clipping in either the likelihood or the predictor. The improvement uses a standard uniform bound for smooth losses. It has the same sample exponent as the elementary clipped argument, at the cost of additional logarithms.

We retain the architecture, energy Q_N, prior, likelihood, and risks from Section 2. There are H hidden layers and one linear readout. All labels lie in $[-1, 1]$, and the joint law of inputs and labels is otherwise arbitrary. A teacher g of width m and weights bounded by B has clones of energy at most $E = d(d+1)B^2 + 2Hm^2B^2$ at every learner width $N \geq m$, and satisfies $|g| \leq U = mB$.

A.1 The smooth-loss inequality

We use the following form of Theorem 1 of Srebro, Sridharan, and Tewari [3, Theorem 1, p. 4 of the full-length version]. If a nonnegative loss is s-smooth in its real-valued prediction and is bounded by b on a function class $\mathcal{F}$, then, with probability at least $1-\eta$, simultaneously for $f \in \mathcal{F}$,

$$R(f) - \widehat{R}_S(f) \le C_0 \Bigg[\sqrt{\widehat{R}_S(f)} \left(\sqrt{s}\log^{3/2}(n)\mathfrak{R}_n(\mathcal{F}) + \sqrt{\frac{b\log(1/\eta)}{n}} \right) + s\log^3(n)\mathfrak{R}_n(\mathcal{F})^2 + \frac{b\log(1/\eta)}{n} \Bigg]. \tag{50}$$

Here $\mathfrak{R}_n$ is the worst-case Rademacher complexity and C_0 is universal; $C_0 = 10^5$ is sufficient. This result assumes neither a convex hypothesis class nor a realizable data law. In particular, it does not assume a Bernstein condition on excess loss relative to the teacher.

The raw square loss is nonnegative and 2-smooth. On $\mathcal{Z}_K$ from (6), its range is bounded by $(\sqrt{K}+1)^2$, since $|z_w| \le \sqrt{Q_N(w)}$. The class $\mathcal{Z}_K$ is symmetric under readout sign reversal, so its Rademacher complexity with an absolute value inside the supremum, as used in [3], equals the convention used in Lemma 2.2.

A.2 The theorem for two or more hidden layers

For $n \ge 2$, $0 < \delta < 1$, and $H \ge 2$, define

$$\begin{aligned} p = H/2, \qquad J_n = \lceil \log_2 n \rceil, \qquad t_n = \log\frac{2(J_n+1)}{\delta}, \qquad q_n = t_n/n, \\ A_n = \frac{\sqrt{2}\, D_n \log^{3/2}(2n)}{H^p\sqrt{n}}, \qquad a_n = 2^p A_n, \\ r_n = C_0(A_n + 2\sqrt{q_n} + A_n^2 + 4q_n). \end{aligned} \tag{51}$$

For a fixed teacher g, put

$$b_n(g,\delta) = (U+1)\sqrt{\frac{2R(g)\log(2/\delta)}{n}} + \frac{2(U+1)^2\log(2/\delta)}{3n}. \tag{52}$$

The slightly conservative coefficient $2/3$ makes this a direct consequence of the usual Bernstein tail inequality.

Theorem A.1 (A raw-output oracle inequality at every learner width)**.** *Let $H \ge 2$, and suppose the likelihood parameter λ satisfies*

$$\frac{1}{n} \le \lambda \le 1, \qquad \lambda^p \ge 8C_0 a_n, \qquad \lambda \ge 32 C_0 q_n. \tag{53}$$

For every fixed teacher g as above, with probability at least $1-\delta$ over the sample, simultaneously for every $N \ge m$,

$$\limsup_{\sigma\downarrow 0} \int R(z_w)\, \Pi_{\sigma,N,S}(dw) \le R(g) + \lambda E + r_n + \frac{8C_0^2 q_n}{\lambda} + b_n(g,\delta). \tag{54}$$

The same right-hand side bounds the limiting risk of the ordinary posterior mean $\int z_w\, d\Pi_{\sigma,N,S}$. It also bounds the limiting risk after clipping that mean, and the limiting risk of the posterior mean of clipped outputs.

An explicit choice is

$$\lambda_n = \max\{(8C_0 a_n)^{1/p},\ 32C_0 q_n\}, \tag{55}$$

for all sufficiently large n that $\lambda_n \le 1$. At fixed d, H, δ, this gives an excess-risk bound of order $\widetilde{O}(n^{-1/H})$ for every fixed teacher. More explicitly,

$$\lambda_n = O_H\left[\left(\frac{D_n^2 \log^3(2n)}{n}\right)^{1/H} + \frac{\log(\log(2n)/\delta)}{n}\right].$$

The tilde suppresses logarithms; it does not suppress a learner-width factor. There is no dependence on N in the displayed bound.

Proof. Apply (50) to each deterministic energy class $\mathcal{Z}_{2^j}$, $0 \le j \le J_n$, using failure probability $\delta/[2(J_n+1)]$. A union bound gives one event of probability at least $1-\delta/2$ for all these classes and all learner widths.

Let w minimize $\widehat{R}_S(z_w) + \lambda Q_N(w)$. Comparing with the zero network gives

$$\widehat{R}_S(z_w) + \lambda Q_N(w) \le \widehat{R}_S(0) \le 1. \tag{56}$$

Write $k = Q_N(w)$. If $k \le 1$, the event for $\mathcal{Z}_1$ implies $R(z_w) - \widehat{R}_S(z_w) \le r_n$. If $k > 1$, choose j with $2^{j-1} < k \le 2^j$. This is possible because $k \le 1/\lambda \le n$. The shell radius $K = 2^j$ satisfies $K \le 2k$. The loss bound obeys $(\sqrt{K}+1)^2 \le 4K \le 8k$. Substituting Lemma 2.2 and $\widehat{R}_S(z_w) \le 1$ into (50) yields

$$R(z_w) - \widehat{R}_S(z_w) \le C_0\left[a_n k^p + a_n^2 k^{2p} + 2\sqrt{2}\sqrt{kq_n} + 8kq_n\right]. \tag{57}$$

Since $p \ge 1$, $k \le 1/\lambda$, and $\lambda^p \ge 8C_0 a_n$, the first two terms satisfy

$$C_0 a_n k^p \le \frac{\lambda k}{8}, \qquad C_0 a_n^2 k^{2p} \le \frac{\lambda k}{64C_0} \le \frac{\lambda k}{64}.$$

The remaining terms obey

$$8C_0 k q_n \le \frac{\lambda k}{4}, \qquad 2\sqrt{2}C_0\sqrt{kq_n} \le \frac{\lambda k}{4} + \frac{8C_0^2 q_n}{\lambda}.$$

Consequently both the $k \le 1$ and $k > 1$ cases imply

$$R(z_w) \le \widehat{R}_S(z_w) + \lambda Q_N(w) + r_n + \frac{8C_0^2 q_n}{\lambda}. \tag{58}$$

Compare the penalized empirical objective with the exact teacher clone:

$$\widehat{R}_S(z_w) + \lambda Q_N(w) \le \widehat{R}_S(g) + \lambda E.$$

The teacher's loss lies in $[0, (U+1)^2]$ and has variance at most $(U+1)^2 R(g)$. Bernstein's inequality therefore gives $\widehat{R}_S(g) \le R(g) + b_n(g,\delta)$ with probability at least $1-\delta/2$. Intersecting the two events proves the right-hand side of (54) for every global minimizer, at every width $N \ge m$.

For completeness, the passage to the posterior also holds for raw risk. At fixed n, N, S, the objective is continuous and coercive. Its minimizer set is compact, and its near-minimum sublevel sets shrink to that set. The risk $w \mapsto R(z_w)$ is continuous on every compact parameter set by dominated convergence, using $|z_w| \le \sqrt{Q_N(w)}$. Moreover,

$$R(z_w) \le 2Q_N(w) + 2.$$

To make the required tail control explicit, write $F(w) = Q_N(w) + \lambda^{-1}\widehat{R}_S(z_w)$ and $F_* = \min_w F(w)$. For every $\eta > 0$,

$$\lim_{\sigma\downarrow 0} \int (1 + Q_N(w))\, \mathbf{1}_{\{F(w) > F_* + \eta\}}\, \Pi_{\sigma,N,S}(dw) = 0. \tag{59}$$

Indeed, a fixed ball around a minimizer lower-bounds the normalizer by its volume times $\exp[-(F_* + \eta/4)/(2\sigma^2)]$. Put $\rho = \eta/[4(F_* + \eta)]$. On the indicated tail, $F \geq (1-\rho)(F_* + \eta) + \rho Q_N = F_* + 3\eta/4 + \rho Q_N$. The remaining integral is a quadratic Gaussian integral of $1 + Q_N$, which grows or decays at most polynomially in σ and is dominated by the factor $\exp[-\eta/(4\sigma^2)]$ after division by the normalizer. This proves (59). Consequently the limiting posterior average of the risk is at most its maximum over the minimizer set, which proves (54). Jensen's inequality and Tonelli's theorem give the conclusion for the ordinary posterior mean. Clipping can only decrease squared loss for labels in $[-1, 1]$; applying the same facts gives the other prediction conventions.

Finally, (55) satisfies $\lambda_n \geq 1/n$ and the other lower bounds in (53). Its leading power is $n^{-1/H}$. The terms r_n have powers at least $n^{-1/2}$, up to logarithms, and q_n/λ_n has power at least $n^{-1+1/H}$. Both are no slower than $n^{-1/H}$ when $H \geq 2$. □

The teacher in this theorem may be selected using the data law and n, but must be fixed before the sample is drawn. In particular, it may minimize the population risk over a specified compact class of bounded-weight teachers. Such a class contains the zero network, so its minimizer has risk at most one. This interpretation gives a sharp oracle inequality with leading coefficient one, rather than a realizability guarantee. The event is simultaneous over learner widths; no simultaneity over all teachers is needed for this interpretation.

A.3 One hidden layer

The same smooth-loss argument gives the rate $\widetilde{O}(n^{-1/2})$ for raw predictions when $H = 1$. Define $p = 1/2$ and the quantities A_n, a_n, q_n, r_n by (51). The following explicit version records the different absorption step.

Theorem A.2 (Raw predictions with one hidden layer)**.** *Let $H = 1$ and suppose*

$$1/n \leq \lambda \leq 1, \qquad \lambda \geq 2C_0(a_n^2 + 8q_n).$$

For every fixed teacher g, with probability at least $1 - \delta$, simultaneously for every $N \geq m$,

$$\limsup_{\sigma\downarrow 0} \int R(z_w)\, \Pi_{\sigma,N,S}(dw) \leq R(g) + \lambda E + r_n + \frac{C_0^2(a_n + 2\sqrt{2q_n})^2}{2\lambda} + b_n(g,\delta). \tag{60}$$

The ordinary posterior mean and the clipped prediction conventions satisfy the same bound. For example,

$$\lambda_n = 32C_0(a_n + \sqrt{q_n} + a_n^2 + q_n)$$

satisfies the lower bounds and gives the rate $\widetilde{O}(n^{-1/2})$ for a fixed teacher, whenever n is sufficiently large that $\lambda_n \leq 1$.

Proof. The same simultaneous shell event yields (57). For $p = 1/2$, its right-hand side is

$$C_0(a_n + 2\sqrt{2q_n})\sqrt{k} + C_0(a_n^2 + 8q_n)k.$$

The second term is at most $\lambda k/2$. Young's inequality bounds the first by

$$\frac{\lambda k}{2} + \frac{C_0^2(a_n + 2\sqrt{2q_n})^2}{2\lambda}.$$

The rest is the teacher comparison, scalar concentration, and Laplace argument from the preceding proof. For the stated schedule, $a_n = \widetilde{O}(n^{-1/2})$ and $q_n = \widetilde{O}(n^{-1})$, giving the claimed rate. □

If a teacher-energy upper bound $E_0 \geq E$ is specified in advance, there is a useful tuning that improves the teacher dependence. Put

$$V_n = \frac{C_0^2(a_n + 2\sqrt{2q_n})^2}{2}, \qquad v_n = \max\{2C_0(a_n^2 + 8q_n),\, 1/n\},$$
$$\lambda_n = \max\left\{\sqrt{\frac{V_n}{E_0+1}},\, v_n\right\}.$$

Whenever $\lambda_n \leq 1$, Theorem A.2 applies, and

$$\lambda_n E + \frac{V_n}{\lambda_n} \leq 2\sqrt{V_n(E_0+1)} + v_n E.$$

At fixed input dimension, this contribution is $\widetilde{O}(\sqrt{(E_0+1)/n} + E_0/n)$. For example, for population-optimal teachers whose width and weights are both at most n^α, one may take $E_0 = O_d(n^{4\alpha})$. The teacher risk is then at most one, and all terms in (60) tend to zero for every $\alpha < 1/4$, with leading power $n^{2\alpha-1/2}$.

References

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[2] Mikael Møller Høgsgaard, Patrick Rebeschini and Tobias Wegel. *Aggregation with Exponential Weights is Optimal in Expectation*. 2026. arXiv:2607.02247.

[3] Nathan Srebro, Karthik Sridharan and Ambuj Tewari. *Smoothness, Low Noise and Fast Rates*. Advances in Neural Information Processing Systems 23, 2010. `https://www.cs.cornell.edu/~sridharan/smoothness.pdf`.